

– gain in kinetic energy of the object

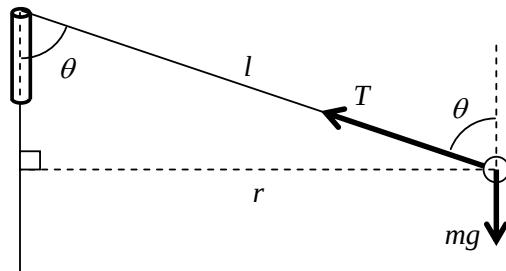
5. **Ans: C**

Output power, $P_o = 0.25P_i = (0.25)(10) = 2.5 \text{ W}$

$$P_o = \frac{E}{t}$$

$$t = \frac{E}{P_o} = \frac{mgh}{P_o} = \frac{(20)(0.50)}{2.5} = 4.0 \text{ s}$$

6. **Ans: D**



For circular motion,

Horizontally, $T \sin \theta = mr\omega^2$

where T = tension in string

θ = angle between string and vertical.

m = mass of bob

r = radius of circular motion.

$$\Rightarrow T \sin \theta = m(l \sin \theta)(2\pi f)^2$$

$$l = \frac{T}{4\pi^2 m f^2}$$

$$l \propto \frac{1}{f^2}$$

Note: T is constant as it is equal to the weight of the brass weights.

